MGT 403 · PROBABILITY MODELING & STATISTICS · YALE SOM · DATA PROJECT

AI prompt log

Where should we open our next preschool? · every prompt we gave the AI, from Problem Set 4 to the final project

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Tool	Claude Code (desktop app), model Claude Opus 5.5, Anthropic
URL	https://claude.ai/code
Dates of access	29–30 September, 6 October, 10–11 October 2026 (per part and section below)
Content	All prompts, verbatim and unedited (typos included), in the order given. No AI output: what the AI produced is in the submitted documents. Attachments are named in brackets. The Census API key and a personal e-mail address are redacted.
Not included	Homework 5, Questions 1–2 (individual homework on stock returns and gelato sales, not part of the data project).

Overview

PART	SCOPE	DATES	PROMPTS	RESULTS IN
1 · Problem Set 4	The distribution of childcare prices: data collection, NDCP prices, CPI, voucher test	29–30 September 2026	23	Submitted PS4 document
2 · Problem Set 5, Question 3	Cost drivers: ACS via Census API, OEWS wages to counties, descriptives, real trends, tests, regression	6 October 2026	21	Submitted PS5 document
3 · Pre-analysis and final project	Pre-analysis tasks 1–4; selection model, checks, slides, appendix	10–11 October 2026	58	Slide deck; Pre-analysis and appendix
Total			**102**	

Prompts marked team's own text contain interpretations the team wrote itself; the submitted versions differ only by spelling and grammar corrections the AI made on request. team's own idea marks the team's guiding ideas for the county selection, which the whole final analysis is built on. team decision marks prompts in which the team chose between options the AI had laid out (the decisions are explained in Appendix D). Each phase of Part 3 names the document where its results appear. Numbering restarts in each part (1.1, 2.1, 3.1, …).

PART 1 · 29–30 SEPTEMBER 2026 · 23 PROMPTS

Problem Set 4

The distribution of childcare prices: data collection, NDCP prices, CPI, voucher test.

Data collection and preparation

1.1 @"C:\Users\phili\Downloads\ASSIGNMENT-updated-2026-09-25.pdf"
We will work on a new project for the Yale course MGT-403 Probability Modeling & Statistics in our current file. For this Project we put ourselves in the shoes of a for profit child care provider. We will be graded on accuracy of our presentation and quality of our reasoning and we want the best garde possible. For this project we will need a lot of data from different sources. I will now give you a document, in which you will look at the data part and pull all the data as described in the document and safe them in a snensible way that we can continue to work with that data. there will be data that i have to manuallyu download try yur best.

1.2 @"C:\Users\phili\Downloads\county_oews_crosswalk.csv" @"C:\Users\phili\Downloads\NDCP2022.xlsx"
Contact email is [e-mail redacted]

1.3 @"C:\Users\phili\Downloads\README.md"
Dont change the source data. If later a grader runs my solution he should use his own email for the contact for download. I have issues with the API Key so we can skip that part for now as we dont need it for Problem 4. Further we have a read me file we should have a look at especially considering the info on the oews data.

1.4 @"C:\Users\phili\Downloads\National-Database-of-Childcare-Prices-Technical-Report.pdf"
Now we should have everything we need for 4.

Question 1

1.5 Now we do Problem Set 4 Subproblem 1. For the NDCP Data we downloaded already we want to restrict that to metro analysis sample, based on the definition counties in an OEWS metropolitan area; see Geography & crosswalk. We will use that for PS4 and PS% and the final Project.

Question 2

1.6 Now we do Problem Set 4 Subproblem 2. We use 2022. Build histograms of median prices by child age group; report mean / SD / number of observations of prices. Compute 95% CIs for the
average value within each age group. Show me the results.

1.7 I will write the discussion myself, as this is my input show me the plots and all the info for that

1.8 team's own text OK I will now write my disscusion in the chat, dont change it only check spelling and grammar.

The 2022 data shows a clear age based price bias across metropolitan counties. The prices decrease as the children get older, meaning infants are the most expensive overall, with an average weekly price of $214.95, followed by toddler care at $186.46 and preschool care at $175.09. This means infant care costs about $40 more per week than preschool care on average. But the variance follows the same trend, with infant prices having the highest SD.
All 3 graphs were skewed to the right, which probably means that high price countries pull up the average.
The 95% confidence interval for the infant mean is [$209.84, $220.07], compared with [$182.36, $190.55] for toddlers and [$171.39, $178.79] for preschoolers. These are quite narrow intervals, with now overlap.
Overall infant care stands out as the most costly, and has the highest geographical variation.

1.9 you can give feedback and what is the issue with bias?

1.10 team's own text The 2022 data shows a clear age based price change across metropolitan counties. The prices decrease as the children get older, meaning infants are the most expensive overall, with an average weekly price of $214.95, followed by toddler care at $186.46 and preschool care at $175.09. This means infant care costs about $40 more per week than preschool care on average. A probable explanation could be that infants require much more stuff to take care of them successfully.
Also variance follows the same trend, with infant prices having the highest SD.
All 3 graphs were skewed to the right, which probably means that high price countries pull up the average.
The 95% confidence interval for the infant mean is [$209.84, $220.07], compared with [$182.36, $190.55] for toddlers and [$171.39, $178.79] for preschoolers. These are quite narrow intervals, which means our average prices are estimated precisely.
Overall infant care stands out as the most costly, and has the highest geographical variation.

That is my solution

Question 3

1.11 Now we do Problem Set 4 Subproblem 3. Convert all dollar values to constant 2022 dollars using CPI-U (CUUR0000SA0; link above).
Plot trends in median price by year for each age group.

1.12 team's own text The key takeaway here is that childcare prices were essentially flat between 2015 and 2022. The nominal price increases, closely match the inflation during this period. If we now look closer within the period there are some changes observable. From 2015 to 2019 the real childcare prices were increasing, meaning that relative to consumer prices, childcare got more expensive during that period. Overall the distribution of prices we saw in Q2 remained the same with infants still being the most expensive. The 2021 dip in prices could be due to data missing from some probably more expensive counties.

1.13 team's own text The key takeaway here is that childcare prices were essentially flat between 2015 and 2022. The nominal price increases, closely match the inflation during this period. If we now look closer within the period there are some changes observable. From 2015 to 2019 the real childcare prices were increasing, meaning that relative to consumer prices, childcare got more expensive during that period. From 2019 to 2022 inflation jumped up, and childcare prices did not keep up. Overall the ramking of prices we saw in Q2 remained the same with infants still being the most expensive. The 2021 dip in prices could be due to data missing from some probably more expensive counties. improve grammar and spelling and add to solution.

Question 4 (my calculations are handwritten, scanned in the main document)

1.14 As I have to do some manual calculation for question 4 i will ad my handwritten notes next.

1.15 for 2022 preschool give me sample mean sample sd sample size so i can do 4 b

1.16 for c I have p= 2*(1-phi(6.38)) calculate result

1.17 how do i unblock to run normal

1.18 disable

1.19 its off rerun

1.20 so when you did the exact calculation with your model you got a different slightly higher p

1.21 repeat the test using 2015 data and subsidy levels

1.22 for c i mistakenly said 6.38 i rounded wrong it is 6.39 give me new p

Submission

1.23 I will now give you my handwritten solution for Problem Set 4 question 4. We now want to prepare the final submission document for Question 2 of Homework 4 for MGT 403 Probability Modeling and Statistics. It is at the same time Problem Set 4 – The Distribution of Childcare Prices of our Data Project. We will combine all the inputs with my handwritten results for question 4 into one clean submission ready PDF document that will achieve the best possible grade. This document will be read by a grader. This prompt will be the last recorded in the prompt log. The prompt log will be part of the appendix of this document. Clean up the prompt log that it is submission ready and only shows my inputs for you that the grader gets a clear idea of the use of ai for this document. The goal is to highlight my manual inputs, like my written texts, ans my handwritten work and combine it with the appropriate AI generated data to answer the questions in the best way possible that they are easy to grade and achieve the highest grade possible. the ammount of good work put into this solution has to be clear. For my handwritten texts in question 4 type them up to increase readability and input them after the handwritten part. for the last part in d regarding 2025 data, supplement my answer with the numbers your gave me to answer it that everything is clear. take your time this has to be submission ready.

@"C:\Users\phili\OneDrive\Dokumente\Stats Group Work Problem Set 4 final.pdf"

PART 2 · 6 OCTOBER 2026 · 21 PROMPTS

Problem Set 5, Question 3

Cost drivers: ACS via Census API, OEWS wages to counties, descriptives, real trends, tests, regression.

2.1 ok now we do question 3 also problem set 5 from the project. My API Key is [REDACTED]. Also keep the prompt log for this task. As before we go step by step with my inputs.

2.2 Lets do task 1: Use the Census API to download ACS 5-year estimates for the explanatory variables listed above, at the county level, for 2015–2022.

2.3 Do we already have the BLS OEWS files to download childcare-worker (SOC 39-9011) wage earnings at the area level, 2015–2022 for question 2. Or didnt we do that already in PS4

2.4 Lets do task 3: Merge OEWS area wages down to counties with the provided county oews crosswalk.csv (join on [year, oews area]: zero-pad (i.e., add leading zeros to) the OEWS area field in the BLS data and transform it to a seven-character variable first so that it matches the crosswalk format), attach ACS county data, and label metros with the area title field from the same OEWS crosswalk. The result should be identified by year × county. Keep only counties in an OEWS metropolitan area (oews area begins 00; see the note above). Sanity-check that nearly all counties have non-missing wages (BLS suppresses a small number of area-year cells, so expect roughly 1% missing) – if many are missing, verify that the merge used the OEWS crosswalk correctly. And explain what you do

2.5 Now we want to build a table for task 4: Build a descriptive table (displaying mean / SD / number of observations) of each predictor in the most recent year (2022).

2.6 Now we do task 5: Convert dollar values to 2022 dollars (CPI-U) and plot trends in childcare-worker wages and rents over time. Plot them in a pretty way

2.7 Next we want to do Task6. For task 6 i have this handwritten input. This has to be part of the documentation in my own words as I again have my manual handwritten ideas that have to be reflected for the best grade possibel: We want to test whether average inflation adjusted childcare worker wages were the same in 2019 and 2022. So we create H0: mu 2022 = mu 2019, which means mu 2022 - mu 2019. That means our H alternative: mu 2022 – mu 2019 unequal to 0. H0 means no real wage change. Therefore we want to run a two sided hypothesis test.

2.8 I forgot "which means mu 2022 - mu 2019 = 0" add in my handwritten part then i will choose the sppropiuate test in my own words

2.9 team's own text My input: I chose a two sided test done at the metro are level. We want to match the metro areas to another for this test, so that the same areas are compared for 2019 and 2022. That allows us to better test H0 and see if the inflation adjusted wages were the same. The data is measured at the metro level. If we were to choose county we would have multiple counties with the same wage level. We want to stay as close to the structure behind the original data as possible.

run the test according to the task

2.10 fix spelling mistake

2.11 team's own text Now my final input for 6: The p value of the test is 0.000011, which is much smaller than the alpha of 5%. Therefore, we reject H0. The results show clear evidence that the childcare workers' wages changed over this period. The averages increased from 2019 $27207.57 to 2022 $27704.73, meaning the change is an increase. That supports the providers claim, but the increased wages do not have to be the reason for increased costs following the Covid-19 pandemic. Further, the increase was relatively small.

2.12 fix spelling and quatify the change in percent add to text in brackets

2.13 Now we do task 7: Bivariate regression: using 2022 data, regress childcare-worker wages on the share of women 25+ with a bachelor's degree or higher (female college share). Report the coefficient. Can you reject that it equals zero? Interpret the coefficient in words. The last two questions will again do in my own words

2.14 what is the difference between the versions

2.15 team's own text First input for 7, we would show the results of the regression after.: I use a bivariate OLS regression at the county level with the standard errors clustered by metro area. Female college share is measured at the county level, whereas the childcare workers' wages are measured at the metro level. Counties within the same metro have the same wage. Claude suggested to then cluster the SE at the metro area to account for this, which I agree with. Therfore Y = a+bX+e, with Y=childcare worker wage and X= female college share.

2.16 we could show the three outcomes do explain why i even have to choose, but we do what creates the best solution for the task

2.17 yes and we should also show the visual

2.18 team's own text input: For the Hypothesis test we want to reject that there is no relationship between the two, so H0: b=0 and HA: b unequal 0.

2.19 ok write we want to test

2.20 team's own text last input We have p<0.0001 meaning with alpha = 5% we can reject H0. There is strong evidence that the female college share is different to 0. The coefficient is 16,499.99, and female college share is measured from 0 to 1. Therefore, a 10 increase in the share of women age 25 and older with a bachelor's degree or higher is associated with $1,650 higher annual childcare-worker wages, on average.

2.21 ok fix both

PART 3 · 10–11 OCTOBER 2026 · 58 PROMPTS

Pre-analysis and final project

Pre-analysis tasks 1–4; selection model, checks, slides, appendix.

Setting up the prompt log 10 October 2026

Results in: This prompt log

3.1 in the end we want a combined prompt log for all parts of the project

3.2 yes keep it uniform we follow the idea we had for the first parts

Pre-analysis: tasks 1–4 10 October 2026

Results in: Pre-analysis and appendix, Part I (tasks 1–4)

3.3 now we start the pre work. Merge NDCP county prices to the year × county cost/demand file from Problem Set 5

3.4 Then define the analysis dataset as year × county observations where prices and predictor variables are non-missing

3.5 Now we ant to do task 2: Make a table of mean / SD / number of observations for all variables in the analysis dataset for 2022.

3.6 Now 3. Make scatter plots of childcare prices against (a) childcare-worker wages and (b) rents, with the fitted line from a linear regression.

3.7 team's own text my input:
Both plots show a strong relationship between childcare prices and operating cost.
Childcare worker wage
Counties with higher childcare workers wages tend to have higher weekly preschool prices. A 1000$ increase in annual worker wadges corresponds with a 10.15$ increase in weekly childcare prices. The R2 indicates that the wgaes explain 54.7% of the variation in childcare prices.
Rent
Counties with higher median monthly rents tend to have higher childcare prices. A 100$ increasre in monthly rent results in about 12.05$ increase in weekly childcare prices. The R2 indicates that rent alone explains 46% of the variation in childcare prices.

Wages have a stronger relationship with childcare prices then rent, based on the respective R2. The two plots show significant variation around the line, meaning that even with similar costs the childcare prices can vary. That is especially visible in the vertical clusters around for example 35000$. The findings suggest both are important predictors of childcare prices, but there are other factors at play.

3.8 fix all

3.9 Now we do Task 4 and we want to do a Multivariate regression: using 2022 data, regress median preschool price on (a) wages alone, (b) rent alone, and (c) wages and rent together.

3.10 team decision 1 we want it to be readblae, 2. we stay metro clusterd for the same reasons as in 5

3.11 do we have all the content for the intepret

3.12 show me the graph

3.13 team's own text my input:
The regressions shows that both childcare worker wages and rent have a positive and significant relationship with weekly preschool prices.
In "a" a 1000$ increase in annual worker wadges corresponds with a 10.15$ increase in weekly childcare prices
In "b" a 100$ increasre in monthly rent results in about 12.05$ increase in weekly childcare prices.
In "c" when wages and rent are both included their coefficient become smaller compared to their individual coefficients. Holding rent constant, a $1,000 increase in annual wages is associated with a $7.09 increase in weekly preschool prices. Similarly, holding wages constant, a $100 increase in monthly rent is associated with a $5.91 increase in weekly prices.
Both variables are positively coreelated with r=0.67, meaning that counties with higher wages also tend to have higher rents. The R^2 also increases to 0.608 in model (c), meaning that wages and rent together explain about 61% of the variation in childcare prices. This is an improvement over either model alone, although approximately 39% of the variation remains unexplained.

3.14 include explanation that was what i ment

3.15 do the fixes

3.16 write model "c"

3.17 write model "c" in the text instead of model(c)

3.18 ok the task ask to Submit a document with answers to these questions and the work underlying them along with your
final project analysis. create a first draft of this document

3.19 the decision is important but we dont need it written out with my words just mention that is what we wanted

Final project analysis: model, selection, reliability, stability, missing factors 10 October 2026

Results in: Slides 1–6; Pre-analysis and appendix, Part II (Appendix 0, A–F)

3.20 lets start with the final project

3.21 *Attachment: image of the "How You Will Be Graded" table: data construction; analysis & method; recommendation & business reasoning; communication*
lets start with the final project. First keep in mind. we want the best possible grade so this hsould guide the project.

3.22 team's own idea we want to find the top 10 most attractive counties for our business. we wan to find counties with unusually high prices relative to costs like wages and rent, prioritize places with strong childcare demand, and identify counties where the advantage has lasted over time. These are our guiding ideas for our selection. we have demand variables from ACS we can include if we need to, to decide between different attractive counties. Then we want to test our approach with the historical data we have to build the best predictive model possible. Lets start with top 20 and then introduce the additional factors to get the top 10. take your time this has to be the best model possible we want the top grade.

3.23	team decision	we use LLS within state with national premium as robustness
3.24	team decision	do the suggestions, metro concentrations are fine we want the best counties
3.25	team decision	no design stuff yet we want to get the math right first
3.26	team decision	rule plus honest mention
3.27		ok lets do d as best as possible
3.28	team decision	do option 1 but we have to explain that decision
3.29		lets do e as best as possible
3.30		is there anything in the analysis we should improve
3.31		do that
3.32		You will be evaluated on accuracy, clarity of thought, and persuasiveness. All data work must make clear what samples and variables were used and what analyses were conducted.

Presentation: story, visuals and slide design 10 October 2026

Results in: Slide deck (structure, figures, design)

3.33		For the slides keep the startimng point from model c in mind and also we want less technical popele to undertsnad waht we did aswell as proffesors in statistics
3.34		lets think about the visualizations first, it has to be the ebst possible for our presentation and done in a stylish way that works best with our data

3.35

The goal is to create a polished, professional presentation that can be delivered in approximately 10 minutes. The main focus should be on clearly communicating the logic, structure, and reasoning behind our analysis, rather than simply presenting statistical results.
We want to tell a compelling story, starting with the preliminary work we conducted, explaining how we built upon that foundation, and highlighting the additional analyses and methodological decisions we made to develop the most accurate and reliable prediction possible.
Audience and accessibility
The presentation should be understandable to an audience with varying levels of statistical knowledge. Even those without a strong statistical background should be able to follow our approach, understand why we made certain decisions, and appreciate the overall logic of our analysis, even if they do not fully understand every statistical technique.
At the same time, the presentation will also be evaluated by our statistics professor and graders. Therefore, it needs to demonstrate academic rigor, methodological soundness, and a solid understanding of the statistical concepts involved.
One important consideration is that some of the statistical methods suggested by the AI model may go beyond what we have covered in class. While we have studied techniques such as multivariate regression, I am not entirely sure whether all the methods included in our analysis fall within the scope of the course. This is not necessarily a problem, but we should be mindful of it. We need to ensure that the methods are appropriate, that we can explain their purpose, and that we do not overcomplicate the main presentation with techniques we cannot confidently defend.
Presentation style and design
Since this is for a Yale School of Management course, the visual style should reflect a high standard of professionalism. I envision something inspired by a McKinsey-style consulting presentation: clean, structured, visually compelling, and focused on communicating key insights.
However, it should not feel overly corporate or generic. We can make it slightly more modern, creative, and visually engaging while maintaining the credibility and sophistication expected in an academic setting.
If additional visual assets, such as the Yale SOM logo, relevant photographs, or other graphics, would improve the presentation, please identify what is needed. I understand that the AI environment may not be able to download images directly from the internet, so I can provide these assets manually.
Main presentation vs. appendix
We should make deliberate use of the appendix, particularly because there is no page limit for it.
The main presentation should remain focused, concise, and engaging enough to deliver comfortably within 10 minutes. It should emphasize the most important insights, key methodological decisions, and overall narrative rather than overwhelming the audience with statistical detail.
The appendix, on the other hand, should contain the deeper technical analysis, additional statistical results, robustness checks, methodological explanations, and supporting evidence. Importantly, the appendix should be just as professionally designed and thoughtfully structured as the main slides, rather than serving as a collection of raw outputs.
Two use cases: live presentation and independent evaluation
A particularly important requirement is that the slides must work equally well in two different contexts:

1. Live presentation: The slides should support a clear, engaging 10-minute explanation in front of the class. They should be visually intuitive, avoid excessive text, and help the audience follow our reasoning.
2. Independent review: The slides must also be sufficiently self-explanatory for the professor and graders to understand our approach, methodology, findings, and conclusions without hearing us present. The appendix should provide the additional depth needed for a thorough academic evaluation.

Overall objective
Ultimately, I want a presentation that combines the storytelling and visual clarity of a top-tier consulting deck with the analytical rigor of a strong graduate-level statistics project.
The audience should walk away understanding not only what we predicted and how well our model performed, but also how we approached the problem, why we made our methodological choices, what we added beyond the initial analysis, and what makes our final prediction credible.

The result should be polished, compelling, academically defensible, and genuinely presentation-ready, with a strong narrative throughout both the main deck and the appendix.

3.36 @"C:\Users\phili\Downloads\MGT_403_2026_Syllabus.pdf"

3.37 the ai stuuf is not important for this project though as acn be seen in the project main file in the geinning all ai use is allowed here

3.38 team decision we can go beyond the classes on the slide but we need to keep that in mind

3.39 *Attachment: image: Yale School of Management logo* ok we want the look to be more modern a bit more apple. and also here the logo

3.40 if needed we can put more on the slides. It is a statistics course after all. we want the best possible grade for our great solution.

3.41 *Attachment: image of the final-project task: ≤ 6 slides / 10 minutes; (a) top 10 listed and visualized, optionally other promising counties; (b) methodology and reliability; (c) market characteristics; (d) stability of above-prediction prices; (e) missing factors; evaluation on accuracy, clarity, persuasiveness; samples, variables and analyses must be clear; appendix document on data handling decisions*
That is the task doe we fufill that task the best way possible

3.42 for other optimal we can show them as a second color on the little map. For the market characteristics we can do a great visualization with gradients or smth, thick about it. speaker notes and appendix later

3.43 team decision the title slide is not important. rather have one more content slide

3.44 Overall can we improve the shade of blues to be more appealing without loosing to much readability. Slide 1: Other promising is overlapping at some points don't like that make the map better. also the how to read box is not necessary, maybe something else better, because we already have the explanations in the graphs. Slide 2: out of sample test we will mention in presentation notes but done need on slide. Think if there is info we miss. Slide 3: Love the funnel. Can we visually improve the market characteristics part. Slide 5: The graph on the left is grade but all the graphs mash into each other, maybe different colors to improve readability Slide 6 maybe a bit more context for the table

Presentation: design refinements 11 October 2026

Results in: Slide deck (design)

3.45 the main blue shade is to bright for me i would want to use a more muted shade of blue as a base. show me some examples

3.46 lets do c

3.47 ok i prefer the version before lets revert

3.48 but i dont like the blue on slide 4. use a more muted one

3.49 ok it is imporant that our intuition in choosing the top 10 gets very clear how we used the dataset and what we analyzed how. is there stuff we should improve to make that even clearer. if so give me options so i can decide

3.50 do 1 do 2 render 3,4 and show me first then we decide

3.51 team decision do 3 not 4

3.52 Consider all the design knowledge and standards you know or research more and see how to improve the presentation to the highest standards. consider symmetries and how people see the slide. there are alignment issues and boxes that can be filled better. the goals is to have a modern corporate presentation that meets all standards. it has to be impeccable and visually appealing. the content is great but we can improve a lot. take your time and show me your improvements and then we decide

3.53 team decision v2 is top

Number check, public pre-K check, appendix 11 October 2026

Results in: Slides (number corrections, public pre-K); Pre-analysis and appendix, Part II

3.54 ignore the speaker notes. check the numbers. check the public k and then do the appendix. the appendix has to be great. all our ideas and intuitions have to lead that appendix. the grader will look at that so take your time and make it perfect

Submission documents 11 October 2026

Results in: Pre-analysis and appendix (combined document, title pages)

3.55 ok how do we want to submit our solution according to the requirents. check all before we proceed.

3.56 what is with the problem sets 4 and 5

3.57 team decision ok lets do pre analysis and appendix in one document in our style

3.58 create title pages for both and insert so that they are clearly seperated